CreditWise: An Intelligent Machine Learning-Based Loan Approval Prediction

Overview

CreditWise is a machine learning-based loan approval prediction system developed for SecureTrust Bank. The system assists in deciding whether a loan application should be approved or rejected based on an applicant's financial and demographic information.

The bank offers home loans and personal loans and faces two major challenges:

1. Good customers are sometimes rejected, resulting in lost business opportunities.
   - Goal: Minimize False Negatives
   - Focus Metric: Recall
2. High-risk customers are sometimes approved, resulting in financial losses.
   - Goal: Minimize False Positives
   - Focus Metric: Precision

To address these challenges, multiple machine learning models were trained and evaluated using both the original dataset and a feature-engineered dataset.

Business Problem

Loan approval decisions directly impact profitability and risk management. The objective is to develop an intelligent classification system that can accurately identify eligible borrowers while minimizing the approval of high-risk applicants.

Machine Learning Models

The following classification algorithms were implemented and compared:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Naive Bayes

Data Preprocessing Pipeline

1. Applicant ID Processing

- Fixed missing Applicant IDs

- Established a clean sequence
- Removed Applicant ID from training data since it does not contribute to prediction

2. Missing Value Treatment

- Median imputation for skewed numerical features
- Mean imputation for symmetric numerical features

3. Exploratory Data Analysis (EDA)

Performed:

- Target variable distribution analysis
- Categorical feature analysis using count plots
- Numerical feature analysis using histograms
- Numerical features vs target analysis using box plots
- Correlation analysis using heatmaps
- Outlier detection

No significant outliers were identified in the dataset.

4. Categorical Encoding

- Label Encoding for ordinal features
- One-Hot Encoding for nominal features

5. Train-Test Split

Dataset divided into training and testing sets.

6. Feature Scaling

Applied feature scaling before model training, particularly important for KNN.

Model A: Baseline Dataset

Features:

- Original cleaned dataset
- Total Features: 28

Workflow:

1. Data preprocessing

2. Model training
3. Performance evaluation

Model B: Feature-Engineered Dataset

Features:

- Original features
- Engineered features
- Total Features: 35

Workflow:

1. Create new domain-specific features
2. Train machine learning models
3. Compare performance with baseline models

Evaluation Metrics

The following metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Special emphasis was placed on:

- Recall to reduce rejection of good customers
- Precision to reduce approval of risky customers

Results

Baseline Dataset (Model A)

Best Performing Model:

- Naive Bayes

- Precision: 80.35%

Feature-Engineered Dataset (Model B)

Best Performing Model:

- Logistic Regression

Key Improvements:

Logistic Regression

- Recall improved by approximately 7%
- Precision maintained at approximately 78%

This indicates better identification of eligible borrowers without sacrificing approval quality.

Naive Bayes

- Precision improved by approximately 1%
- Recall decreased by approximately 4%

This indicates a slightly more conservative approval strategy.

Key Findings

- Feature engineering significantly improved Logistic Regression performance.
- Logistic Regression achieved the best balance between business objectives.
- Higher recall reduced the rejection of qualified applicants.
- Precision remained competitive, helping control lending risk.
- KNN showed comparatively lower effectiveness as the feature space increased.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Future Enhancements

- Hyperparameter tuning using Grid Search
- Ensemble learning methods
- XGBoost and Random Forest implementation
- Explainable AI techniques using SHAP and LIME
- Deployment using Flask or Streamlit

Conclusion

CreditWise demonstrates how machine learning can support intelligent loan approval decisions by balancing business growth and risk management. Feature engineering substantially improved model performance, with Logistic Regression emerging as the most suitable model for SecureTrust Bank's loan approval process.